

415 Final Project

Neo Kok

2023-11-13

Data

```
X_train <- read.csv("data/X_train.csv", header=TRUE)
y_train <- read.csv("data/y_train.csv", header=TRUE)
y_sample <- read.csv("data/y_sample.csv", header=TRUE)
X_test <- read.csv("data/X_test.csv", header=TRUE)
```

Feature engineering

```
# Create average of SRP variable
X_train$average_srp <- rowMeans(X_train[, grepl("^SRP", names(X_train))], na.rm = TRUE)

# Factorize district, self_eval, and teacher_eval
X_train$district <- as.factor(X_train$district)
X_train$self_eval <- as.factor(X_train$self_eval)
X_train$teacher_eval <- as.factor(X_train$teacher_eval)
X_train$y = y_train$y
```

Model Creation

```
# Create multiple variable linear model with interaction term
lm = lm(y_train$y ~ self_eval * district + teacher_eval + average_srp, data = X_train)
```

Model Evaluation

```
summary(lm)$r.squared
```

```
## [1] 0.8944066
```

Model Predictions

```

# Feature engineering for test sample
X_test$average_srp <- rowMeans(X_test[, grepl("^SRP", names(X_test))], na.rm = TRUE)
X_test$district <- as.factor(X_test$district)
X_test$self_eval <- as.factor(X_test$self_eval)
X_test$teacher_eval <- as.factor(X_test$teacher_eval)

# Predict test sample using model
predictions <- predict(lm, newdata = X_test)
y_test = data.frame(SEQN = X_test$SEQN, y = predictions)

# Save out predictions
#write.csv(y_test, "y_test.csv", row.names = FALSE)

```